Norm.

Prop. Let $f \in C([0,1])$. Then,

$$\lim_{n\to\infty}\left[\int_0^1 |f(x)|^n\, dx\right]^{\frac{1}{n}} = \max\{|f(x)| \mid x\in[0,1]\}.$$

Proof. Since $[0,1]$ is compact, the image of $|f|$ is also compact, so for some $a\in[0,1]$, $|f(a)| = \max\{|f(x)| \mid x\in[0,1]\}$. Observe that

$$|f(a)| = (|f(a)|^n)^{\frac{1}{n}} = \left[\int_0^1 |f(a)|^n\, dx\right]^{\frac{1}{n}}$$

Meanwhile, since $[0,1]$ is compact, so f is continuous uniformly on $[0,1]$.
Given $\varepsilon>0$, choose $\delta>0$ such that $|x-y|<\delta \Rightarrow |f(x)-f(y)|<$
By the Minkowski inequality, for each $n\in\mathbb{N}$,

$$\left|\left[\int_0^1 |f(x)|^n\, dx\right]^{\frac{1}{n}} - \left[\int_0^1 |f(a)|^n\, dx\right]^{\frac{1}{n}}\right| \le \left[\int_0^1 |f(x)-f(a)|^n\, dx\right]^{\frac{1}{n}}.$$

$$\|x+y\| \le \|x\| + \|y\|$$

$$\|x+y\| - \|x\| \le \|y\|$$

$$\overset{\shortparallel}{x_i}\quad \|\underbrace{x_i - x_j}_{x} + \underbrace{x_j}_{y}\| - \|\underbrace{x_j}_{y}\| \le \|x_i - x_j\|$$

Therefore, for each $n\in\mathbb{N}$,

Prop. Let $(x_1,\cdots,x_n)\in\mathbb{R}^n$ be given. Then

$$\lim_{p\to\infty}\left[\sum_{i=1}^n |x_i|^p\right]^{\frac{1}{p}} = \max\{|x_1|, |x_2|, \cdots, |x_n|\}$$

Proof. Observe that let $1\le j\le n$ be the index such that

$$|x_j| = \max\{|x_1|, \cdots, |x_n|\}.$$

Put $(y_i)_{i=1}^n = (x_j, x_j, \cdots, x_j)$.
Then $(y_i)_{i=1}^n$ satisfies that

$$\|(y_i)\|_p = \left[\sum_{i=1}^n |x_j|^p\right]^{\frac{1}{p}} = (n\cdot|x_j|^p)^{\frac{1}{p}} = \sqrt[p]{n}\cdot|x_j| \to |x_j| \text{ as } p\to\infty.$$

Now, by the Minkowski Inequality,

$$\|(y_i)\|_p - \|(x_i)\|_p \le \|(y_i)-(x_i)\|_p = \|(x_j - x_i)_{i=1}^n\|_p = \left[\sum_{i=1}^n |x_j - x_i|^p\right]^{\frac{1}{p}}$$